

Required speed and torque

1. FBD

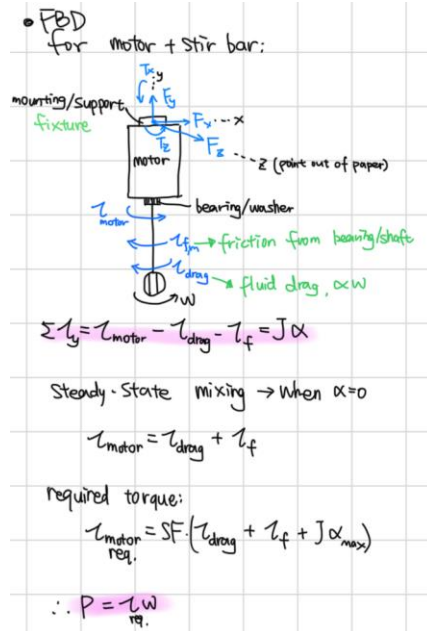


Figure 1: free-body diagram of motor and stir bar

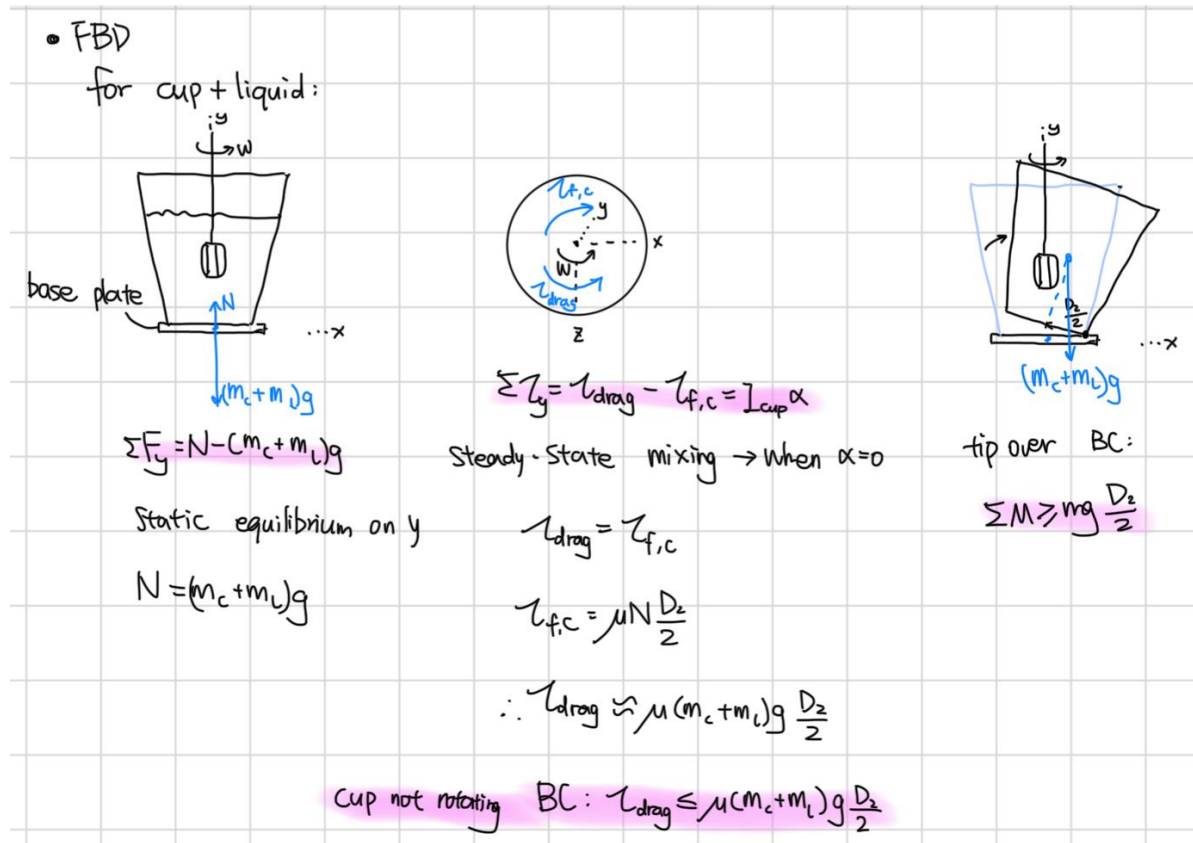


Figure 2: free-body diagram of cup and liquid

2. Estimate dimensions, parameters, and assumptions

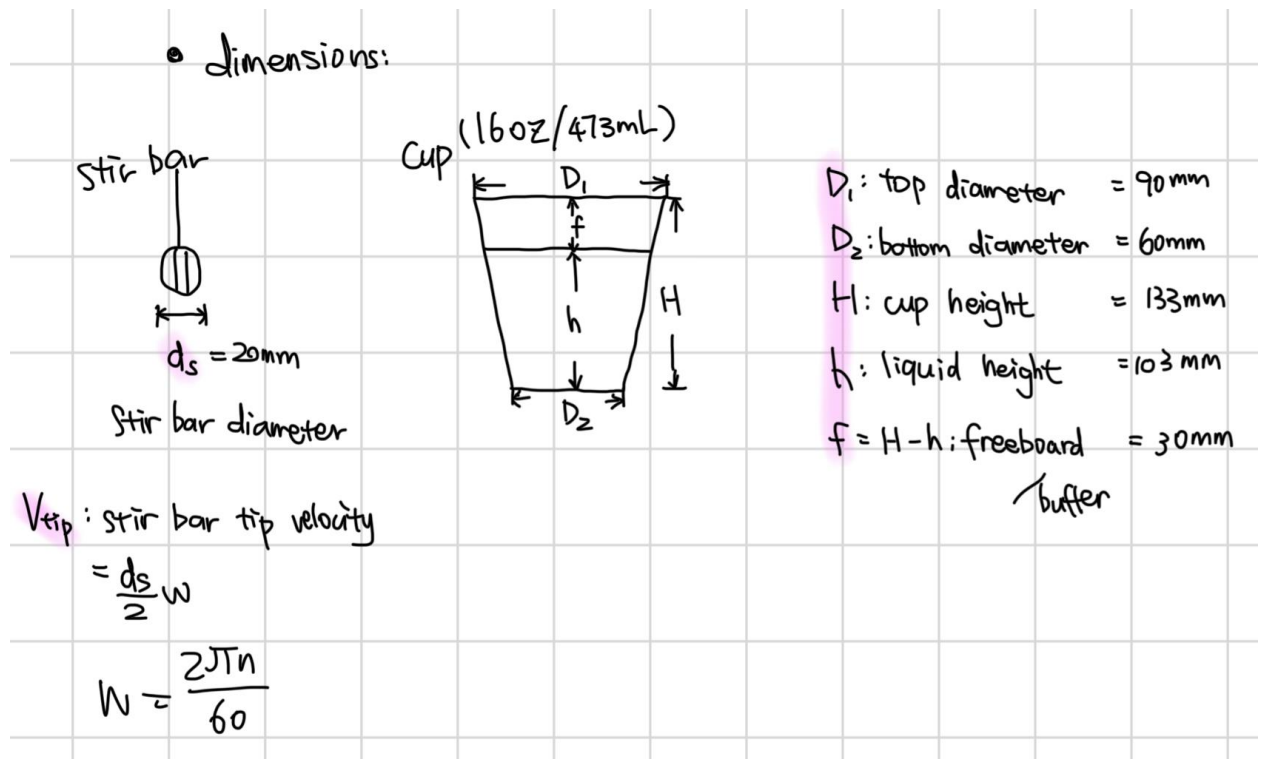


Figure 3: dimension schematics of stir bar and cup

- Fill up 80% volume of the cup, liquid volume is fixed to be 16oz/473mL. ($h = 0.8H$)
- Paper/plastic cup mass $m_c = 20\text{g}$
- Average liquid density $\rho = 1050\text{kg/m}^3$
- Mixing time $t = 15\text{s}$
- diameter of stir bar $d_s = 20\text{mm}$
- Height of stir bar $h_s = 10\text{mm}$
- linear speed of stir bar tip $v_{tip} = 1\text{m/s}$

3. Speed range

- Too low: Produce weak bulk circulation, longer mixing time
- Too high: Cup tipping over, spilling/splashing, creating too much foam

Thus, set our estimate mixing time to be 15s, and determine our lower speed range by:

$$Re = \frac{\rho n d_s^2}{\mu}$$

Where:

Re = Reynolds number

ρ = density of liquid [kg/m³]

n = rotational speed [rpm]

μ = dynamic viscosity ≈ 0.001 Pa · s for water-based liquid

If we set lower rotational speed to be $n = 300$ rpm,

$$Re = \frac{\rho n d_s^2}{\mu} = \frac{(1050)(300)(0.02)^2}{0.001} \approx 2100$$

For stirred tank systems, $Re > 2000$ typically indicates turbulent or transitional mixing, which enhances bulk circulation.

To determine the higher range of rotational speed, we use tip speed and splash boundary conditions, set v_{tip} to be 1m/s.

$$\omega = \frac{2\pi n}{60}$$

ω = angular speed [rad/s]

n = rotational speed [rpm]

$$v_{tip} = \frac{d_s}{2} \omega = \frac{d_s}{2} \left(\frac{2\pi n}{60} \right)$$

$$n = \frac{2v_{tip}}{d_s} \left(\frac{60}{2\pi} \right) = \frac{2(1)}{\frac{20}{1000}} \cdot \left(\frac{60}{2\pi} \right) \approx 955rpm$$

To provide margin/buffer for freeboard variation and vibration during demonstration, the operating maximum speed was set to 800 rpm.

Thus, the speed range of the motor would be 300-800 rpm, default/avg $n = 500$ rpm.

4. Torque and power

- Liquid viscosity range - τ_{drag}
- mixing waiting time

Assumptions: mixing is 15s at 300rpm, to determine drag torque applied on stir bar by liquid, using dynamic pressure equation:

$$q = \frac{1}{2} \rho v_{tip}^2 = \frac{1}{2} \rho \left(\frac{d_s}{2} \frac{2\pi n}{60} \right)^2 = \frac{1}{2} (1050) \left(\frac{0.02}{2} \frac{2\pi(300)}{60} \right)^2 \approx 51.48 \text{ pa}$$

$$\tau_{drag} = FL = qA \frac{d_s}{2} = q(d_s h_s) \frac{d_s}{2} = (51.48)(0.02)(0.01) \left(\frac{0.02}{2} \right) \approx 0.00010296 \text{ N} \cdot \text{m}$$

The dynamic pressure method underestimates drag because it neglects vortex formation and bulk circulation. Therefore, a conservative engineering estimate safety factor of 3 was selected based on similar small beverage mixers.

$$\tau_{drag} = 3 \cdot (0.00010296) \approx 0.000286 \text{ N} \cdot \text{m}$$

In small DC motor systems with light radial loading, bearing friction torque is typically on the order of 10–50% of applied load torque. A mid-range value estimate was selected as a conservative design assumption.

$$\tau_{f,m} \approx 0.01 \text{ N} \cdot \text{m}$$

As motor runs, rotational speed gradually increases by constant angular acceleration until steady-mixing state and reach defined rpm. Let's set accelerating/deaccelerating time to be 2s each, we got our angular velocity vs time plot and acceleration vs time plot:

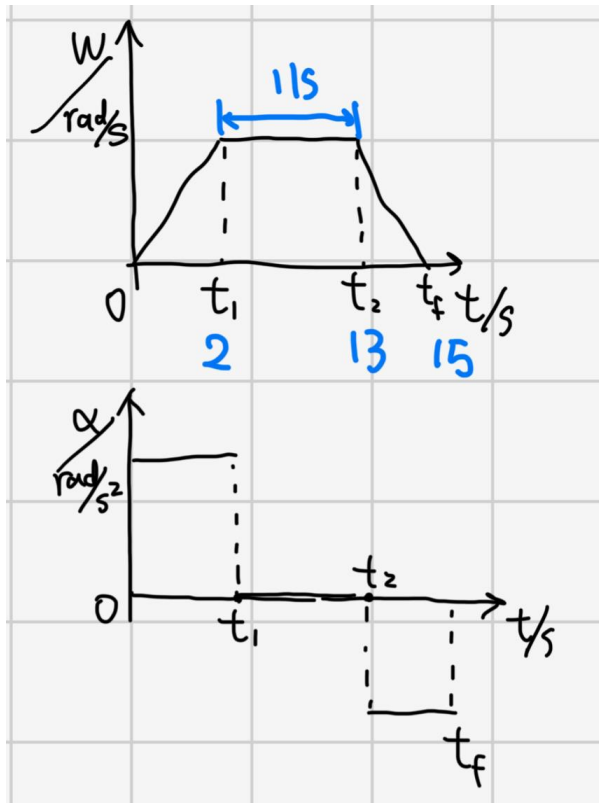


Figure 4: angular velocity vs time plot and acceleration vs time plot

To find maximum acceleration,

$$\alpha_{max} = \frac{\Delta\omega}{t_1} = \frac{\frac{2\pi n}{60} - 0}{2} = \frac{2\pi(300)}{60(2)} \approx 15.708 \frac{rad}{s^2}$$

$$J = J_{shaft} + J_{stir\ bar}$$

Let's assume that stir bar is solid disk, shaft is slender rod, bearings is cylinder.

Shaft dimensions: length $l = 80\text{mm}$, diameter $d = 3\text{mm}$, density of stainless steel $\rho = 8000\text{kg/m}^3$

$$m = \rho V = \rho \pi \left(\frac{d}{2}\right)^2 l = (8000)\pi \left(\frac{0.003}{2}\right)^2 0.08 = 4.5\text{g}$$

$$J_{shaft} = \frac{1}{2} m \left(\frac{d}{2}\right)^2 = \frac{1}{2} (4.5)(0.001) \left(\frac{0.003}{2}\right)^2 \approx 5 \times 10^{-9} \text{ kg} \cdot \text{m}^2$$

Stir bar mass $m = 3\text{g}$:

$$J_{\text{stir bar}} = \frac{1}{2}m\left(\frac{d}{2}\right)^2 = \frac{1}{2}(3)(0.001)\left(\frac{0.02}{2}\right)^2 \approx 1.5 \times 10^{-7} \text{ kg} \cdot \text{m}^2$$

Thus,

$$J = J_{\text{shaft}} + J_{\text{stir bar}} = 5 \times 10^{-9} + 1.5 \times 10^{-7} \approx 1.6 \times 10^{-7} \text{ kg} \cdot \text{m}^2$$

$$\tau_{\text{motor}} = SF(\tau_{\text{drag}} + \tau_{f,m} + J\alpha) = 2(0.000286 + 0.01 + (1.6 \times 10^{-7})(15.708)) \\ \approx 0.01029 \text{ N} \cdot \text{m}$$

$$P = \tau_{\text{motor}} \cdot \omega = (0.01029)\left(\frac{2\pi(300)}{60}\right) = 0.64629 \text{ W}$$

Reference:

1. [Amazon.com: Nahida Black Milk Frother Wand for Coffee, Frother Handheld Rechargeable Electric Whisk Drink Mixer with 3 Heads 3 Speeds Coffee Frother Electric Stirrer For Latte, Cappuccino, Hot Chocolate: Home & Kitchen](#)
2. [Amazon.com: YUSWKO Beige Handheld Milk Frother Wand for Coffee, Rechargeable Electric Whisk with 3 Heads 3 Speeds Drink Mixer Coffee Frother For Latte, Cappuccino, Hot Chocolate, Egg: Home & Kitchen](#)
3. [Amazon.com: Aphse Milk Frother Handheld Electric Milk Foamer Portable Mini Mixer Machine Stainless Steel Touch Handheld Espresso Mixer Battery Operated Frother Coffee Stirrer Maker for Hot Chocolate Coffee: Home & Kitchen](#)